

PATARA TRIRAT

AI Research Scientist
KAIST Ph.D.

Affiliation: Principal AI Research Scientist at DeepAuto.ai

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RESEARCH INTERESTS

As an AI research scientist, I am passionate about advancing agent-based task automation toward human-agent collaboration and exploring new frontiers of perception in autonomous agents operating in complex, real-world environments through multimodal large language models. My research interests include

Agentic AI & LLM Agents: TS-Debate [[arXiv](#)], AutoML-Agent [[Paper](#)], Agentic Predictor [[Paper](#)], MONAQ [[Paper](#)], ZeroPOIRec [[Paper](#)], OmniRetrieval [[arXiv](#)]

Spatiotemporal Data Mining: DF-TAR [[Paper](#)], MG-TAR [[Paper](#)], Universal Time Series Survey [[arXiv](#)], Time-PEFT [[Paper](#)], MacroLens [[arXiv](#)]

AutoML: AutoML-Agent [[Paper](#)], Conversational AutoML [[Paper](#)], PASTA [[Paper](#)], TFAS [[Paper](#)]

EDUCATION

AUGUST 2020 — FEBRUARY 2025 / DAEJEON, SOUTH KOREA

KAIST

Ph.D in School of Computing, GPA: 3.99 / 4.30

Dissertation: Neural Architecture Search for Time-Series Analysis with Context-Specific Performance Estimators

Advisor: Prof. Jae-Gil Lee

Undergraduate Student Research Mentoring:

- + Phatarapran Saraluck
- + Wirittipol Supdateyarnnakorn

Additional Activity:

- + **KAIST Student Ambassador:** KAIST Office of Admissions
- + **AI Engineer Intern:** DeepAuto Co., Ltd.
- + **Club Member:** Lunatic Street Dance Club
- + **Invited Talks:** KSC2021 Top Conference Session and International Seminar on Policies for Older Drivers

AUGUST 2018 — AUGUST 2020 / DAEJEON, SOUTH KOREA

KAIST

M.S. in Knowledge Service Engineering (Currently, Data Science), GPA: 3.80 / 4.30

Thesis: Analysis of Dangerous Driving Behavior from Driving Log Data for Traffic Accident Risk Prediction

Advisor: Prof. Jae-Gil Lee

Additional Activity:

- + **Student Assistant Photographer:** KAIST International Scholar & Student Services (ISSS)
- + **Club Member:** Lunatic Street Dance Club

JUNE 2012 - MAY 2016 / BANGKOK, THAILAND

Kasetsart University

B.S. in Computer Science with First Class Honors, GPA: 3.74 / 4.00

Advisor: Prof. Pakaket Wattuya

Additional Activity:

- + **Web Developer** (Co-operative Education): NewKru Co., Ltd. (Supervisor: Prof. Sukumal Kitisin and Prof. Usa Sammapun)
- + **Teaching Assistant:** Web Technology and Web Services (CS443)
- + **Dance Instructor / Club Member:** KU Dance Club

EMPLOYMENT

NOVEMBER 2025 — PRESENT / SEOUL, SOUTH KOREA

Principal AI Research Scientist at **DeepAuto Co., Ltd.**

JANUARY 2025 — OCTOBER 2025 / SEOUL, SOUTH KOREA

Senior AI Research Scientist at **DeepAuto Co., Ltd.**

Multi-Agent System Development, Automated Design of Multimodal Agentic AI Systems, Tool Generation, LLM Agents for Time-Series Understanding

JANUARY 2024 — MAY 2024 / SEOUL, SOUTH KOREA

AI Engineer Intern at **DeepAuto Co., Ltd.**

Automated Machine Learning (AutoML), Large Language Model for AutoML using Multi-Agent Framework

AUGUST 2018 — JANUARY 2025 / DAEJEON, SOUTH KOREA

Research Assistant at **Data Mining Lab, KAIST**

Spatial-Temporal Data Mining and Analysis, Deep Learning Research, Automated Machine Learning for Time Series and Data Stream

MAY 2017 — AUGUST 2018 / BANGKOK, THAILAND

Frontend Developer at **Sobtid Edutainment Co., Ltd.**

Built APIs and developed web applications under sobtid.me project (both services for customers and in-house officers) using Node.js on Amazon's AWS and the web application using React.js with Google's Firebase

NOVEMBER 2016 — FEBRUARY 2017 / BANGKOK, THAILAND

Intern at **Korean Cultural Center**

Advertisement Design, Document Translation, and Event Organization Assistance

JUNE 2015 — NOVEMBER 2015 / BANGKOK, THAILAND

Web Developer at **NewKru Co., Ltd.**

Researched on MOOC platforms and developed a new MOOC application

PUBLICATIONS

Preprint / Ongoing Work

- + Trirat, P., Kwak J.M., Heo J., Lee H., and Hwang, S.J.
MacroLens: A Multi-Task Benchmark for Contextual Financial Reasoning under Macroeconomic Scenarios
<https://arxiv.org/abs/2606.24950>
- + Trirat, P., Heo J., Lee H., and Hwang, S.J.
Visual Anchoring for Scenario-Guided Forecasting
Ongoing Work
- + Trirat, P., Ha, J., Joo, H., Heo J., Lee H., and Hwang, S.J.
Proactive Problem Solver: LLM Agents for the Enterprise R&D Pipeline
Ongoing Work
- + Trirat, P., Zhang, S., Heo J., and Hwang, S.J.
Agentic Tool Generation with Hierarchical Goal-Driven Co-Planning
Ongoing Work
- + Trirat, P., Kwak J.M., Heo J., Lee H., and Hwang, S.J.
TS-Debate: Multimodal Collaborative Debate for Zero-Shot Time Series Reasoning
<https://arxiv.org/abs/2601.19151>
- + Baek, J., Jeong, S., Park, S., Yeo, W., Kang, M., Trirat, P., Lee, H., Hwang, S.J.
OmniRetrieval: Unified Retrieval across Heterogeneous Knowledge Sources
<https://arxiv.org/abs/2605.29250>

Conference Proceedings

- + Na, J., Trirat, P., Park, C., and Lee, J.
Time-PEFT: Temporal and Multichannel Complexity-Based Fine-Tuning for Time-Series Foundation Models
ICML 2026 (CORE A*)
- + Trirat, P., Jeong, W., and Hwang, S.J.
Multi-View Encoders for Performance Prediction in LLM-Based Agentic Workflows
ICLR 2026 (CORE A*), ICML 2025 MAS Workshop (Outstanding Paper)
- + Trirat, P. and Lee, J.
MONAQ: Multi-Objective Neural Architecture Querying for Time-Series Analysis on Resource-Constrained Devices
EMNLP 2025 (CORE A*, Findings)
- + Trirat, P., Jeong, W., and Hwang, S.J.
AutoML-Agent: A Multi-Agent LLM Framework for Full-Pipeline AutoML
ICML 2025 (CORE A*)
- + Nam, Y., Trirat, P., Kim, T., Lee, Y., and Lee, J.
Context-Aware Deep Time-Series Decomposition for Anomaly Detection in Businesses
ECML PKDD 2023 (CORE A, Applied Data Science, Oral Presentation)
- + Na, J., Kim, J., Trirat, P., Park, J., Park, S., Yu, K., and Lee, J.
Interpretable Relation Prediction of Undirected Hypergraphs with Dynamic Bipartite Graphs
KIISE, KCC 2023
- + Trirat, P.*, Nam, Y.*, Kim, T., and Lee, J.
AnoViz: A Visual Inspection Tool of Anomalies in Multivariate Time Series
AAAI 2023 (CORE A*, Demonstration Paper)

- + Trirat, P., Shin, Y., Kim, S., and Kim, M.
Generating a Machine Learning Model with a Few Sentences
KIISE, KSC 2021
- + Trirat, P. and Lee, J.
DF-TAR: A Deep Fusion Network for Citywide Traffic Accident Risk Prediction with Dangerous Driving Behavior
TheWebConf 2021 (CORE A*, Oral Presentation)
- + Trirat, P., Noree, S., and Yi, M.Y.
IntelliMOOC: Intelligent Online Learning Framework for MOOC Platforms
EDM 2020 (CORE B, Poster Paper)
- + Trirat, P., Kim, M., and Lee, J.
Experimental Analysis of the Effect of Autoencoder Architectures on Time Series Representation Learning
KIISE, KCC 2020
- + Shin, Y., Yoon, S., Trirat, P., and Lee, J.
CEP-Wizard: Automatic Deployment of Distributed Complex Event Processing
IEEE ICDE 2019 (CORE A*, Demonstration Paper)

Journal Articles

- + Trirat, P., Shin, Y., Kang, J., Nam, Y., Na, J., Bae, M., Kim, J., Kim, B., and Lee, J.
Universal Time-Series Representation Learning: A Survey
ACM Computing Surveys (2026) Vol. 58 No. 12 (SCIE, Impact Factor: 30.4)
- + Kim, J., Seo, Y., Kim, Y., Kang, J., Shin, J., Trirat, P., and Lee, J.
Large Language Models are Zero-Shot Point-of-Interest Recommenders
Data Mining and Knowledge Discovery (SCIE, Impact Factor: 5.5, ECML PKDD 2025 Journal Track, Oral Presentation)
- + Trirat, P. and Lee, J.
TFAS: Zero-Shot NAS for General Time-Series Analysis with Time-Frequency Aware Scoring
Machine Learning (SCIE, Impact Factor: 4.9, ECML PKDD 2025 Journal Track, Oral Presentation)
- + Trirat, P. and Lee, J.
PASTA: Neural Architecture Search for Anomaly Detection in Multivariate Time Series
IEEE Transactions on Emerging Topics in Computational Intelligence (SCIE, Impact Factor: 6.5)
- + Trirat, P., Yoon, S., and Lee, J.,
MG-TAR: Multi-view Graph Convolutional Networks for Traffic Accident Risk Prediction
IEEE Transactions on Intelligent Transportation Systems (2023) Vol. 24 No. 4 (SCIE, Impact Factor: 9.1)
- + Trirat, P., Yoon, S., and Lee, J.,
Spatial and Temporal Analysis of Dangerous Driving Behavior Using DTG Data
Korea Database Research (2019) Vol. 35 No. 3 (KCI)
- + Trirat, P., Yoon, S., and Lee, J.,
Study on Driving Behavior of Different Vehicle Types Using DTG Data
Korea Database Research (2019) Vol. 35 No. 2 (KCI)

ACADEMIC SERVICES

Conference Reviewer

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining
- SIAM International Conference on Data Mining
- International Conference on Machine Learning
- International Conference on Learning Representations
- Annual Conference on Neural Information Processing Systems
- AAAI Conference on Artificial Intelligence
- ICML 2025 Workshop on MAS
- ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving
- ACL Rolling Review (ARR): May 2026

Journal Reviewer

- Transactions on Machine Learning Research
- ACM Computing Surveys
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Mobile Computing
- IEEE Transactions on Emerging Topics in Computational Intelligence
- Journal of Supercomputing

Research Mentoring & Collaboration

- Geonju Lee (Research Intern / Master's Student)
- Jisung Ha (Research Intern)
- Jihye Na (Ph.D. Student)

LANGUAGES
AND SKILLS

Languages

- Thai:** Native
English: Effective Operational Proficiency
> **TOEIC:** 975 / 990
Korean: Limited Working Proficiency
> **TOPIK IBT:** 6 / 6
> **TOPIK:** 5 / 6
> **TOPIK Speaking:** 3 / 6

Programming Languages

Python, JavaScript, HTML/CSS, R, Java

Frameworks / Libraries

Tensorflow, Keras, React.js, Node.js, PyTorch

Cloud Services

Google Cloud Platform, Amazon Web Services, Microsoft Azure

Softwares

Microsoft Offices, Apple iWork, Adobe Photoshop, Apple Final Cut Pro

ACCOMPLISHMENTS

Honors and Awards

- + 2026 *Gold Reviewer Award* (Top 25%) by ICML 2026
- + 2025 *Outstanding Paper Award* by ICML 2025 Workshop on Multi-Agent Systems in the Era of Foundation Models: Opportunities, Challenges and Futures
- + 2025 *Commendable Reviewer* by KDD (February Cycle) 2025
- + 2025 *Top 10% Outstanding Reviewer* by KDD (August Cycle) 2025
- + 2023-24 *Student Ambassador* by KAIST Office of Admissions
- + 2023 *University Visit Program Scholarship Recipient* by KAIST School of Computing
- + 2022 외국인 유학생 독도사랑 한국어 말하기대회 최우수상 (팀명: K-GOOSE) by Daegu University
- + 2020-24 *KAIST Scholarship* (Doctoral Program) by KAIST
- + 2019 3rd Place at *Year End Party Talent Show* by KAIST ISSS
- + 2018-20 *KAIST Scholarship* (Master Program) by KAIST

- + 2017 Winner at *Thailand Hip Hop Dance Championship* (Megacrew Division) by Hip Hop International Thailand
- + 2016 Winner *LINE Finance Night Dance Competition* by LINE Finance
- + 2014-16 *Faculty of Science Awards* for Outstanding Academic Achievement Student by Faculty of Science, Kasetsart University
- + 2013-15 *Medal of Excellent Academic Achievement* by Kasetsart University
- + 2013-15 *5As Scholarship* by Department of Computer Science, Kasetsart University
- + 2012-14 *Tuition fee Exemption* (5As+ per semester) by Kasetsart University
- + 2012 *King Rama IX's Royal Awards* (Qualified Grantee of Big Vocational College) by Ministry of Education of Thailand
- + 2011 3rd Runner Up at *MOS Olympics Competition Thailand* by ARiT
- + 2011 Winner H.R.H Princess Soamsawali Trophy at *SIAM MOS Competition* by ARiT

Certificates

- + 2022.08 Letter of Appointment (*Member of Dokdo Supporters*) by Dokdo Foundation
- + 2019.07 Certificate of Completion (*Introduction to Data Science*) by Skooldio
- + 2019.04 Certificate of Completion (*2019 Spring Mentor Program*) by KAIST ISSS
- + 2018.03 *Mobile Sites Certification* by Google
- + 2018.03 Certificate of Completion (*Modern React with Redux*) by Udemy
- + 2017.08 Certificate of Completion (*React Js and Redux Mastering Web Apps*) by Udemy
- + 2017.05 Certificate of Training (*Mr.Wiggles Hip Hop Camp*) by Studio Zoom
- + 2016.08 Certificate of Achievement (*K-Singing Class*) by Korean Cultural Center
- + 2016.04 *Test of Practical Competency in ICT* (TOPCIT — Score: 297, Level: 2) by IITP and KPC
- + 2015.12 Certificate of Completion (제1기 한국유학대비 한국어과정: 중급) by Korean Education Center
- + 2014.05 Certificate of Participation (*Thailand K-POP Camp for Young Talents*) by Korean Cultural Center and Silpakorn University
- + 2014.01 Certificate of Completion (*5th Master Class*) by Move With Passion
- + 2013.10 Certificate of Honor (*Super Start Summer Camp*) by Monkey Town Academy
- + 2013.08 Certificate of Participation (*ACM-ICPC Thailand Central Group A Programming Contest 2013*) by ACM-ICPC
- + 2013.03 Certificate of Training (*1st Software Developing Skills Training Camp*) by Department of Computer Science
- + 2012.10 Certificate of Training (*1st Programming Skills for Problem-solving Training Camp*) by Depart of Computer Science

Major Course Projects at KAIST

- + Tiny-TimeNAS: A Python Package for Time-Series Analysis on IoT Devices with Tiny Neural Architecture Search (CS565: IoT Data Science)
- + A3M: Adaptive Memory Management Mechanism for Online Task-free Continual Learning (AI602: Advanced Deep Learning)
- + Q&AI: AI-Infused Q&A System for Online Learning (CS492 (J): Special Topics in Computer Science<AI for Smart Life>)

- + Towards Better Understanding of Cryptocurrency: A Multi-Aspect Analysis (CS564: Data Science Methodology)
- + Neural Predictor for Neural Architecture Search in Tensorflow 2.0 (CS470: Introduction to Artificial Intelligence)
- + Mr.TAD: Time Series Anomaly Detection with Multi-resolution Ensemble Learning (CS570: Artificial Intelligence and Machine Learning)
- + Conversational AutoML: A Framework for Generating an Efficient Machine Learning Model with a Few Sentences (CS579: Computational Linguistics)
- + HandyBoard: An Adjustable Split of Soft Keyboard in Landscape Mode Smartphone Touch Screens (CS584: Human-Computer Interaction)
- + Jet Presentation: A Presentation Rehearsal Partner for Everyone (KSE624: Mobile and Pervasive Computing for Knowledge Services)
- + TRepNet: Convolutional Denoising Autoencoder for Time Series Representation Learning (KSE801: Data Analysis based on Deep Learning)
- + Answering Question in Massive Open Online Course by Recommending Ranked Video Clips (KSE643: Knowledge Engineering and Intelligent Decision Making)

Previous Projects

- + Korean Square Web Application (React.js)
- + D Maniac Studio Mobile Application (React Native)
- + D Maniac Studio Web Application (React.js)
- + Sneakavilla Online Store (Wordpress)
- + Kyouei Co., Ltd. Website (Wordpress)
- + Sriracha Tugboat Co., Ltd. Website (Wordpress)
- + Web Application UX/UI Design (Human-Computer Interaction Assignment and Project using AngularJs)
- + Robotz Language and Compiler (Formal Language & Automata Theory Project using Java)
- + Hook Flash Card E-Learning Web Application (Web Technology and Web Service Project#2 using CakePHP)
- + Workpoint Web-based Resource Management (Web Technology and Web Service Project#1 using PHP)
- + Floral Bloom (ERP Project using SAP B1)
- + Flower Bloom Online Store (Ruby on Rails on IBM Bluemix)
- + Adventure in Wonderland (Software Engineering Game Project using Java)
- + Hangman Game (Network Application Project using Python with Tk GUI)
- + Index Sequential Search (Data Structure Project using Adobe Flash Animation)
- + Fan Store (Software Construction Project using Java)
- + Electricity Cost Prediction Program (DevCamp#1 Windows 8 Application Project using C#)
- + Hangman CLI Game (AlgoCamp#1 using Python)
- + Polynomial Calculation (Computer Programming Project using Python)
- + Store Project (Computer Programming Project using Python)

On-stage Performances

- + 2025 CLICK Battle Vol. (Top 4)
- + 2023 Lunatic - Spring Performance
- + 2022.11 Lunatic - Funk Night Vol.4 Tier Battle (Top 2 / **Finalist**)
- + 2022.07 1MDJ Showcase - Best Dancer
- + 2022 Lunatic - Spring 오픈동방 Battle (Top 4)
- + 2019 Lunatic - Spring PPOP Battle (Top 8)

- + 2019 Lunatic - Open Battle (Prelims)
- + 2019 Lunatic - 기획부장배 Battle (Top 8)
- + 2019 Lunatic - Spring 오픈동방 Battle (Guest Show)
- + 2018 Lunatic - Fall Freshmen Battle (Top 8)
- + 2018 Lunatic - Fall Illusion x Lunatic Battle (Top 8)
- + 2018 Lunatic - Fall Performance
- + 2017 D Maniac Megacrew - Hip Hop International Thailand (**Winner**)
- + 2017 D Maniqueen - Hip Hop International Thailand (Ranked 12th)
- + 2016 D Maniqueen - LINE Finance Night Dance Competition (**Winner**)
- + 2016 D Maniqueen - Hip Hop International Thailand (Ranked 6th)
- + 2016-18 KU Dance Club - Kaset Fair Annual Festival
- + 2016 KU Dance Club - Kasetsart University Orientation Day
- + 2015 D Maniqueen - Seacon Street Int'l Challenge (Ranked 6th)
- + 2015 D Maniqueen - To Be Number One Variety Showcase (NBT Channel)
- + 2015 TomYumKoong - Hip Hop International Thailand (Ranked 4th)
- + 2015-16 KU Dance Club - KU First Date
- + 2015-16 KU Dance Club - Freshy Night
- + 2014-16 KU Dance Club - Open Activity World Event
- + 2015-16 KU Dance Club - Royal Project Foundation Annual Event at Central World

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OTHER INTERESTS

Art and Sport

Performing Arts, Acting, Dancing, Singing, Table Tennis, Running, Basketball

Music

K-Ballad, K-Pop, Rap/Hip-Hop, R&B, Soundtracks

Lifestyles

Photography, Dramas/Movies, Mainstream Social Medias